Demand Forecasting and Inventory Optimisation

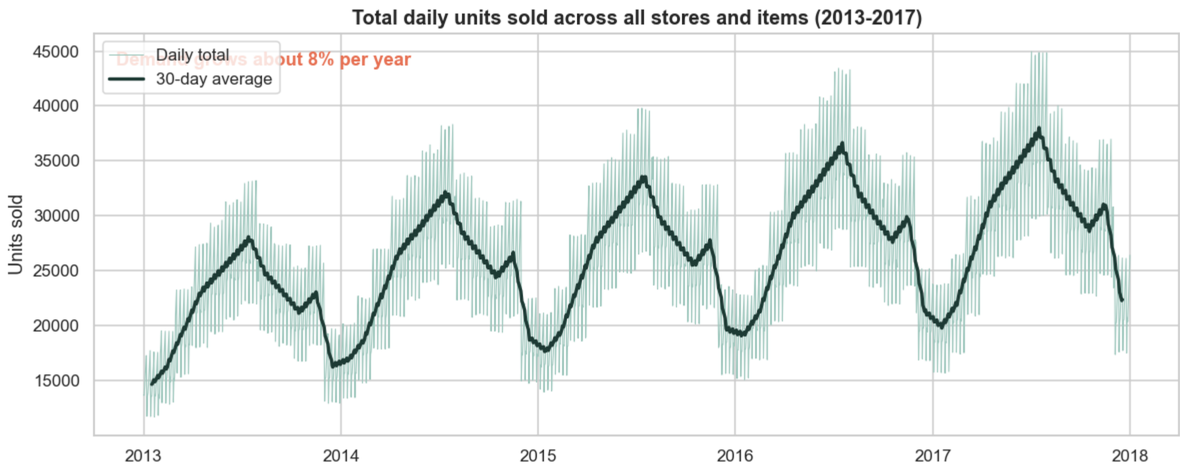
Forecasting daily sales for 500 store-item combinations, then turning the forecast into smarter stocking decisions. Kingsley Amegah, Data Scientist.

913,000	11.4%	about 50%	13%
daily records (2013-2017)	forecast error (WMAPE)	less safety stock	less inventory, same service

1. The business problem

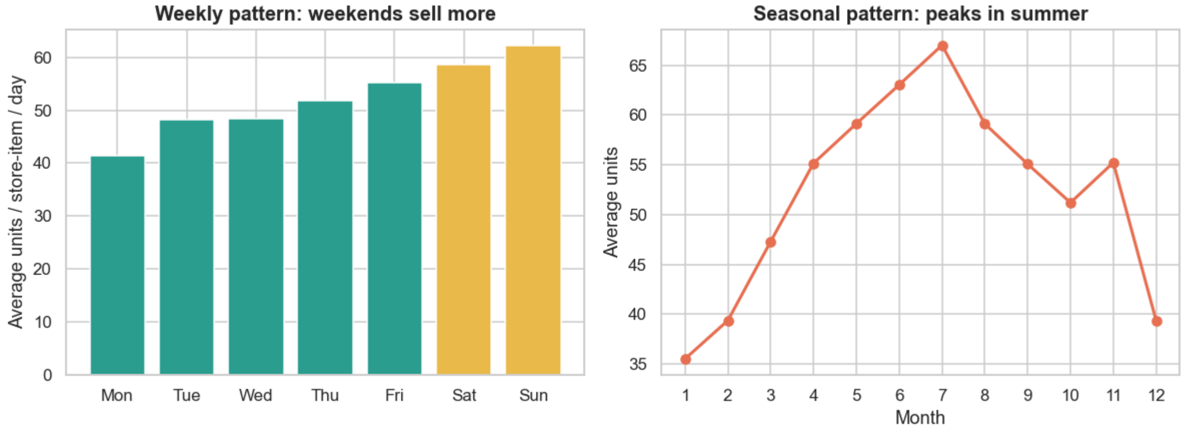
Every retailer faces the same tension. Hold too little stock and you get stockouts, meaning lost sales and unhappy customers. Hold too much and you tie up cash, fill the warehouse and risk markdowns and waste. The lever that controls both is how accurately demand can be predicted, and how well that prediction is turned into reorder decisions. Using five years of real daily sales for 10 stores and 50 items, this project learns the demand patterns, builds and tests a forecasting model, and converts the forecast into an inventory policy whose value is measured in a simulation.

2. How demand behaves



What this shows. Total units sold each day, with a 30-day average. Demand grows about 7.8% per year with a strong yearly wave.

What it means. A forecast must capture the trend and the seasonal shape, not just repeat last week, or it will run low in the busy season and overstock in the quiet one.

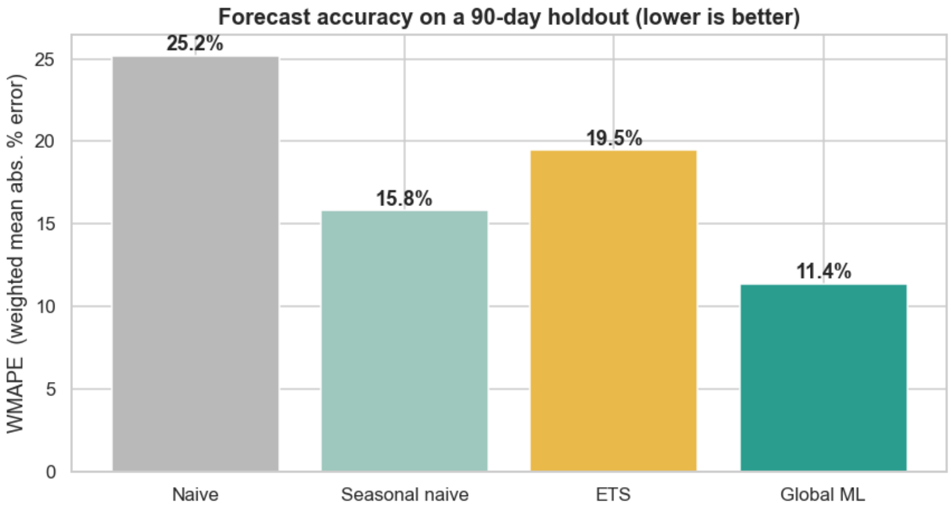


What this shows. The weekly and monthly patterns. Weekends sell about 23.3% more than weekdays and demand peaks in summer.

What it means. These two rhythms are the backbone of the forecast and should guide staffing and deliveries.

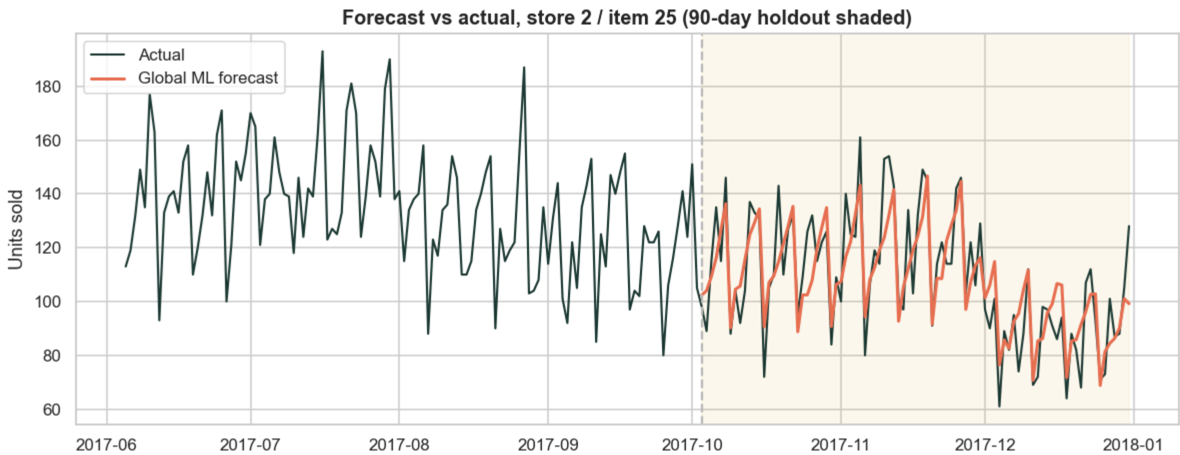
3. Forecasting: testing four models honestly

To trust a forecast it must be tested on unseen data. I held out the last 90 days and compared four models. Every feature the machine-learning model uses is at least 90 days old, so it never sees into the period it predicts.



Model	WMAPE	MAE
Naive	25.21%	12.73
Seasonal naive	15.8%	7.98
ETS	19.46%	9.83
Global ML	**11.39%**	**5.75**

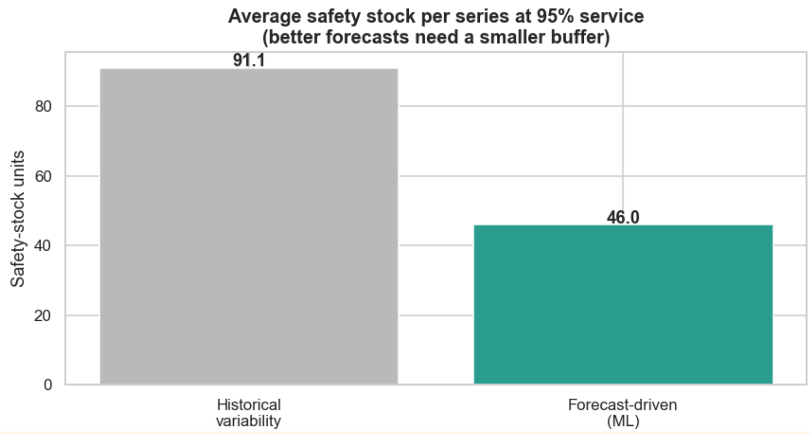
What it means. The machine-learning model wins clearly at 11.39% error, about 28% more accurate than the strong seasonal-naive benchmark. Learning from all 500 series at once lets it borrow patterns a single-series model misses.



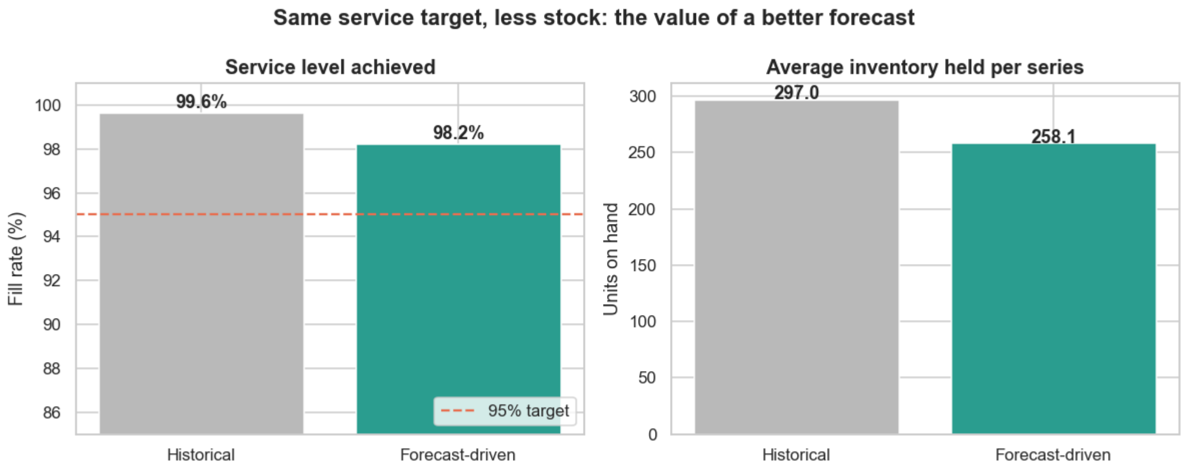
What this shows. One store-item; the shaded area is the 90-day test window the model never saw. The forecast tracks both the weekly swing and the December slowdown.

4. From forecast to inventory decisions

A forecast creates value only when it changes a decision. Each forecast becomes a weekly order-up-to policy: every week, top each store-item up to a level covering expected demand over the reorder cycle plus a safety-stock buffer sized for 95% service. The buffer depends directly on how uncertain the forecast is.

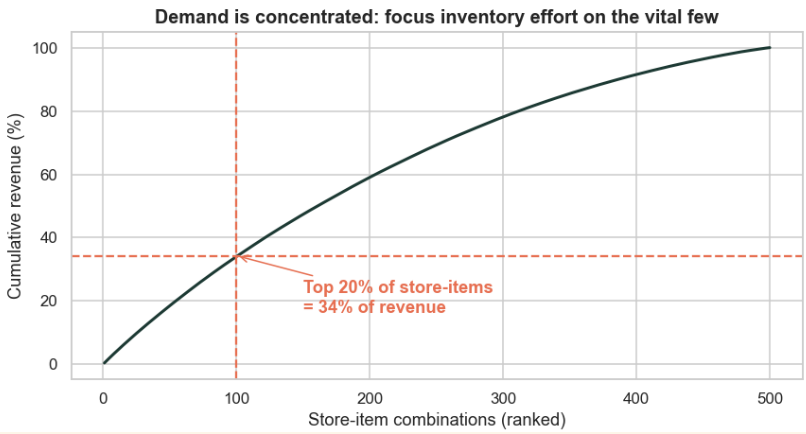


What it means. Because the model already explains the weekly and seasonal swings, the leftover uncertainty is far smaller, so the buffer can be about 50% smaller for the same service promise.



What this shows. A day-by-day simulation of the weekly reorder policy across all 500 store-items, run from historical variability (grey) versus from the forecast (teal).

What it means. Both clear the 95% target, but the forecast-driven policy does it with 13% less inventory on hand. The historical policy reaches 99.7% only by overstocking, capital that earns nothing.



What it means. The top 20% of store-items drive about 34% of revenue, so forecasting and buffering effort should be tightest there.

5. Recommendations

1. Adopt the machine-learning forecast for replenishment, about 28% more accurate than current practice and scalable to all series from one model.

2. Re-size safety stock from forecast error, not raw demand swings, roughly halving the buffer and cutting inventory about 13% while still meeting the 95% service target.

3. Differentiate by importance, holding the tightest service on the top 20% of store-items that drive a third of revenue.

4. Validate live before full rollout, piloting on a sample of stores, tracking realised service and inventory for a quarter, then scaling and re-training as demand shifts.

6. Method and limitations

Data: Store Item Demand Forecasting Challenge (Kaggle), 10 stores, 50 items, daily sales 2013-2017, no missing values. Forecasting: a global HistGradientBoostingRegressor with calendar features and lags of 90+ days, benchmarked against naive, seasonal-naive and ETS on a 90-day holdout. Inventory: a periodic-review order-up-to policy, 7-day lead time, 95% target, evaluated in a day-by-day simulation. The cost figures use assumed unit, holding and order costs; the percentage improvements do not depend on those assumptions. Lead time is treated as fixed; real lead-time variability would call for a slightly larger buffer.

Built in Python (pandas, scikit-learn, statsmodels) from open data. Code: github.com/Kingsley-amg/demand-forecasting